

Piyush Chaudhari

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SUMMARY

I work at the intersection of Statistics and Machine Learning, and their applications in Clinical Informatics, Modeling and Computation, and Data Democratization

RESEARCH INTERESTS

Statistical Machine Learning, Clinical Informatics and Decision Support, Patient Representation Learning, Phenotyping and Sub-Phenotyping, Privacy-Preserving Zero Knowledge Learning, Data Democratization

EDUCATION

Ph.D. in Biomedical Informatics

2025 - current

University of Florida, Gainesville

Master of Science in Biostatistics

2016 - 2018

University of Florida, Gainesville

Bachelor of Technology in Biotechnology

2012 - 2016

Vellore Institute of Technology, India

WORK EXPERIENCE

• Data Management Analyst III at [UF Health Integrated Data Repository](#) (Feb 2023 - Current)

• [UF Health](#)

- Curating data elements from EPIC Clarity and Caboodle and make them available for use in interdisciplinary research projects
- Developing Extract Transform Load (ETL) pipelines to deploy and validate data in Common Data Models like [OMOP](#), [PCORNET](#) and [i2b2](#)

• [UF Clinical and Translational Science Institute](#)

- Designing and developing data science pipelines to improve the coverage and quality of data for UF Health Research Enterprise
- Maintaining databases for free-text sources like clinical notes, order and pathology reports, drivers of health variables, genetic testing data, and additional health data sources
- Liasioning as an honest broker between researchers and UF Health/Shands data server to provision access to data based on IRB approved research protocols

• [UF Cancer Institute](#)

- Leading the data team to build the Cancer Data Commons, on the lines of [NCI Genomic Data Commons](#), with access to multimodal data for cancer patients
- Building domain-specific infrastructure for Cancer with novel extraction methods for [NAACCR Tumor Registry](#) elements and deployment of oncology elements in OMOP CDM

• [i2b2](#) (Informatics for Integrating Biology and the Bedside)

- Leading the development of pipeline from EPIC-native Caboodle Data Model to i2b2 data model
- Building i2b2 running on OMOP Common Data Model
- Implementing and validating i2b2 [Digital Twin Technology](#) based on Computational Phenotypes and Loyalty Cohorts to improve cohort phenotyping
- Maintaining [ENACT Network](#) i2b2 as a UF Health liasion. ENACT Network is comprised on about 59 health system across US

• [OneFlorida+ Clinical Research Consortium](#)

- Leading the transition to OMOP Common Data Model which will improve the workflow of 20+ [OneFL+ partners](#)
- Designing i2b2 application that is built over PCORNET common data model with clinical data of 31 million patients across Florida, Georgia, Alabama, California and Minnesota
- Serving as UF Health honest broker for multi-site studies under [PCORNET Network](#)
- Building novel, first of its kind infrastructure to bring large-scale nursing data into OneFlorida+ clinical research network (PI: [Macieira](#))
 - Grant Support: NIH [R21/R33](#)

• [Bridge2AI: Patient-Focused Collaborative Hospital Repository Uniting Standards \(CHoRUS\) for Equitable AI](#)

- Working as UF Health Clinical Data Expert to deliver AI-ready datasets in OMOP format (Site PI: [Bihorac](#) and [Rashidi](#))
 - Grant Support: NIH [3OT2OD032701-01S3](#)

• Other Selected Projects

- Serving as a concept expert to create a local instance of [BigMouth](#) multi-site dental data repository and i2b2 for [UF College of Dentistry](#)
- Leading the integration of multi-modal data concepts like images, genomic results and waveforms into OMOP CDM

• Selected accomplishments

- **20-25x** improvement in structured data delivery to research customers
- **10x+** improvement in data delivery for free-text clinical notes and order reports
- **5x+** improvement in data delivery through OMOP Common Data Model
- **5x+** improvement for custom de-identification for data delivery

- **Data Management Analyst II** at *UF Department of Epidemiology (Jun 2018-Jan 2023)*
 - Conducted statistical analyses, data management and quality control at [UF HealthStreet](#) (substance abuse and public health epidemiology data) for ~13,000 individuals across 700 data points
 - Created visualizations, hotspots and heatmap analysis for intervention planning
 - Worked with [UF Integrated Data Repository](#) to link self-reported data from community members to their Electronic Health Records at [UF Health](#), analyzing data from [UF Survey Research Center](#), [BEHR](#)
 - Provided technical support for Diagnostic and Statistical Manual of Mental Disorder (DSM-V) software (developed in MS Access) to clients across US
 - Selected Projects (Served as Lead Statistician):
 - **All of Us Consortium of CTSA Community Engagement Programs** (PI: [Cottler](#))
 - Grant Support: National Institute of Health (NIH)
 - **National Drug Early Warning System Coordinating Center (NDEWS)** (PI: [Cottler](#))
 - Grant Support: National Institute on Drug Abuse (NIDA)
 - **Program to Alleviate National Disparities in Ethnic and Minority Immunizations in the Community, Vaccination and Vaccine Education Focus** (PI: [Cottler](#))
 - Grant Support: Center for Disease Control (CDC) Partnering for Vaccine Equity
 - **Women Helping Women- Lung Cancer Screening Study** (PI: [Striley](#))
 - Grant Support: [UF Cancer Institute](#)
 - **Community Engagement and Research Program- HealthStreet** (PI: [Cottler](#))
 - Grant Support: NIH National Center for Advancing Translational Sciences (NCATS)
 - **Precision Public Health Approaches to Reduce Disparities in Memory Disorder Screening in Rural Minority Communities** (PI: [Cottler](#))
 - Grant Support: FL Department of Health
 - **Study of Non-Oral Administration of Prescription Stimulants** (PI: [Cottler](#))
 - Grant Support: [Arbor Pharmaceuticals](#)
- **Graduate Research Statistician** at *UF Department of Biostatistics (Aug 2017-May 2018)*
 - Involved in data reconfiguration, management, and consulting for the [Southeastern and Coastal Center for Agricultural Safety and Health \(SEC-CaSH\)](#)

TEACHING AND MENTORING

- Currently assisting with course teaching and design for [GMS 6803 Data Science for Clinical Research \(Fall 2026\)](#)
 - Course instructors: Drs. [Yi Guo](#) and [Mei Liu](#)
- Conducting [i2b2](#) and [OHDSI ATLAS](#) training workshops for professionally diverse cohorts like medical students, residents, clinical faculty, research staff and students at UF Health and other national research partners from *(Feb 2023 - Current)*
- Mentored for [EGN 4912 Engineering Directed Independent Research](#) undergraduate program *(Spring 2026)*
- Managed statistical design and provided research mentorship for graduate students and postdoctoral fellows working on [Cottlerlab](#) legacy and ongoing research data *(Jun 2018 - Jan 2023)*

TECHNICAL SKILLS

Programming : Python, R, SQL, SAS, Markdown

Common Data Models : OMOP, i2b2, PCORNET, EPIC Clarity, EPIC Caboodle

Softwares and Packages : MS Power BI, SPSS, RedCAP, ArcGIS, Git, SAP Data Services, SAP Business Objects

Core Concepts : Data Visualization, Statistical Modeling and Diagnostics, Statistical Computing, Predictive Analytics, Hashing, Clinical Informatics, Privacy-Preserving Informatics, Data Democratization, Applied Survival Analysis, Linear and Generalized linear models, Linear and Non-Linear Mixed Models, Optimization, Cross Validation, Machine Learning, Natural Language Processing, Simulation studies, Clinical Trial Design, Interactive Matrix Language, Monte Carlo methods

TALKS AND WORKSHOPS

- **Presenter**
 - i2b2 training for [GMS 6847 - Translational Research and Therapeutics: Bench, Bedside, Community, & Policy](#) class *(2023 - current)*
 - i2b2, ATLAS and SlicerDicer training for UF College of Medicine Faculty Bootcamp Session *(2023 - current)*
 - OneFlorida+ i2b2 training for Florida State University and Tallahassee Memorial HealthCare *(Oct 2023)*
 - OneFlorida+ i2b2 training for Advent Health University *(Sep 2023)*
- **Attended**
 - UF Health EPIC Cosmos Datathon *(May 2026)*
 - 2024 OHDSI Global Symposium *(Oct 2024)*

PROFESSIONAL SERVICE

- Volunteer, [AMIA Knowledge Discovery and Data Mining](#) Working group *(July 2026 - current)*

PUBLICATIONS

HealthStreet Cancer Survivor Cohort: A Community Registry for Cancer Research

Apr 2024

Cheng TD, [Chaudhari PV](#), Bitsie KR, Striley CW, Varma DS, Cottler LB. The HealthStreet Cancer Survivor Cohort: a Community Registry for Cancer Research. J Cancer Surviv. 2024;18(2):366-374. <https://doi.org/10.1007/s11764-022-01173-4>

Drug deactivation pouches for primary prevention of opioid overdose: perceptions and attitudes of community members

Mar 2024

Otufowora A, Egan KL, [Chaudhari PV](#), Okusanya AA, Ogidan AO, Cottler LB. Drug Deactivation Pouches for Primary Prevention of Opioid Overdose: Perceptions and Attitudes of Community Members. J Am Board Fam Med. 2024;37(1):112-117. Published 2024 Mar 11. <https://doi.org/10.3122/jabfm.2023.230227R1>

Insomnia among community members in Florida: Associations with demographics, health conditions, and social support

May 2023

Fidler AL, **Chaudhari P**, Sims V, Payne-Murphy J, Fischer J, Cottler LB. Insomnia among community members in Florida: Associations with demographics, health conditions, and social support. J Clin Transl Sci. 2023;7(1):e128. Published 2023 May 5. <https://doi.org/10.1017/cts.2023.536>

Electronic Nicotine Delivery Systems (ENDS) use among members of a community engagement program

Apr 2023

Austin-Datta RJ, **Chaudhari PV**, Cheng TD, Klarenberg G, Striley CW, Cottler LB. Electronic Nicotine Delivery Systems (ENDS) use Among Members of a Community Engagement Program. J Community Health. 2023;48(2):338-346. <https://doi.org/10.1007/s10900-022-01169-2>

CONFERENCE PROCEEDINGS

Social Role Protective Factors in Risky Substance Use Among a Community Sample of Emerging Adults

Jul 2024

Fenton, M.; **Chaudhari, P.**; Cottler, L.; Striley, C. Social Role Protective Factors in Risky Substance Use Among a Community Sample of Emerging Adults. Drug and Alcohol Dependence 2024, 260, 110635. <https://doi.org/10.1016/j.drugalcdep.2023.110635>

Characterizing Chronic Pain And Co-Occurring Substance Use Across The Adult Lifespan In A Florida Community Sample

Apr 2023

Horta, M.; **Chaudhari, P.**; Cottler, L.; Fillingim, R.; Striley, C. Characterizing Chronic Pain And Co-Occurring Substance Use Across The Adult Lifespan In A Florida Community Sample. The Journal of Pain 2023, 24 (4), 111. <https://doi.org/10.1016/j.jpain.2023.02.314>.

COMMUNITY-RECRUITED OLDER ADULTS DIFFER FROM PATIENT POPULATIONS

Dec 2022

Striley C, **Chaudhari P**, Fillingim R, Cottler L. COMMUNITY-RECRUITED OLDER ADULTS DIFFER FROM PATIENT POPULATIONS. Innov Aging. 2022;6(Suppl 1):841-842. Published 2022 Dec 20. <https://doi.org/10.1093/geroni/igac059.3017>

Sustaining community engagement during COVID-19: High stress and loneliness reported

Apr 2022

Striley CW, Varma DS, **Chaudhari PV**, Cottler LB. 160 Sustaining community engagement during COVID-19: High stress and loneliness reported. Journal of Clinical and Translational Science. 2022;6(s1):17-17. <https://doi.org/10.1017/cts.2022.68>

As Food Insecurity Worsens During COVID-19, Negative Mental Health Impact on Community Members Increases

May 2021

Striley C, **Chaudhari P**, Varma D, Cottler L. 44191 As Food Insecurity Worsens During COVID-19, Negative Mental Health Impact on Community Members Increases. Journal of Clinical and Translational Science. 2021;5(s1):144-144. <https://doi.org/10.1017/cts.2021.768>

Service referral follow up rate among participants of a community engagement initiative in Florida

Mar 2019

Varma DS, **Chaudhari PV**, Vaddiparti K, Striley CW, Cottler LB. 3418 Service referral follow up rate among participants of a community engagement initiative in Florida. Journal of Clinical and Translational Science. 2019;3(s1):94-94. <https://doi.org/10.1017/cts.2019.215>

CERTIFICATIONS

- EPIC Cosmos Data Architect (Mar 2026)
- EPIC Clarity (May 2025)
- EPIC Caboodle (May 2025)
- EPIC Cogito (May 2025)
- EPIC Clinical Data Models (May 2025)
- SAS Certified Specialist: Base Programming Using SAS 9.4 (Aug 2021)
- Design for Six Sigma - License: 13361772 (Jun 2018)
- Six Sigma Green Belt Certified - License: 13361771 (Jun 2018)

AWARDS AND SCHOLARSHIPS

ERS PhD Fellow

2025-2026

As a Emerging Research Scholars PhD Fellow for College of Medicine, UF, I will be

- building AI and data science methods to advance health around core principles of community, trustworthiness, and diversity
- supporting the efforts in our community, state, and nationwide as a part of The UF College of Medicine faculty, students, and trainees

University of Florida Superior Accomplishment Award

2025

Recognizes faculty and staff members who contribute outstanding and meritorious service, efficiency and/or economy, or have created an impact on the quality of life provided to students and employees

- University-Gold Award for Individual Employee Performance in Finance, Research, and Compliance
- Division-Level Winner (Division 5- Health Affairs) for Individual Employee Performance

IN MEDIA

- A champion for medical databases. Doctor Gator. (accessed 2026-07-26)
- UF Medicine faculty, staff honored with Superior Accomplishment Awards . Doctor Gator. (accessed 2026-07-26)
- Guest Columnist
 - UF HealthStreet provides nasal spray that reverses opioid overdoses, Gainesville Sun. (accessed 2026-07-26)